3.1 THERMODYNAMICS

Thermodynamics is the study of the energetics of chemical reactions. There are two relevant forms of energy in chemistry: kinetic energy (movement of molecules) and potential energy (energy stored in chemical bonds). [What is the most important potential energy storage molecule in all cells?[1]] The **first law of thermodynamics,** also known as the **law of conservation of energy**, states that the energy of the universe is constant. It implies that when the energy of a system *decreases*, the energy of the rest of the universe (the **surroundings**) must *increase*, and vice versa. The **second law of thermodynamics** states that the disorder, or **entropy**, of the universe tends to increase. Another way to state the second law is as follows: spontaneous reactions tend to increase the disorder of the universe. The symbol for entropy is S, and "a change in entropy" is denoted ΔS, where $\Delta S = S_{\text{final}} - S_{\text{initial}}$. [If the ΔS of a system is negative, has the disorder of that system increased or decreased?[2]]

A practical way to discuss thermodynamics is the mathematical notion of **free energy** (**Gibbs free energy**), defined by Josiah Gibbs as follows:[3]

$$\textbf{Eq. 1} \quad \Delta G = \Delta H - T\Delta S$$

T denotes temperature, and H denotes **enthalpy**, which is defined by another equation:

$$\textbf{Eq. 2} \quad \Delta H = \Delta E + P\Delta V$$

Here E represents the bond energy of products or reactants in a system, P is pressure, and V is volume. [Given that cellular reactions take place in the liquid phase, how is H related to E in a cell?[4]] ΔG increases with increasing ΔH (bond energy) and decreases with increasing entropy.

- Given the second law of thermodynamics and the mathematical definition of ΔG, which reaction will be favorable: one with a decrease in free energy ($\Delta G < 0$) or one with an increase in free energy ($\Delta G > 0$)?[5]

The change in the Gibbs free energy of a reaction determines whether the reaction is favorable (**spontaneous**, ΔG negative) or unfavorable (**nonspontaneous**, ΔG positive). In terms of the generic reaction

$$\text{A} + \text{B} \rightarrow \text{C} + \text{D}$$

the Gibbs free energy change determines whether the reactants (denoted A and B) will stay as they are or be converted to products (C and D).

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1 ATP, which stores energy in the ester bonds between its phosphate groups.

2 If ΔS is negative, then the system lost entropy, which means that disorder decreased.

3 As in ΔS, the Greek letter Δ (delta) indicates "the change in." For example, $\Delta G_{\text{rxn}} = G_{\text{products}} - G_{\text{reactants}}$.

4 $H \approx E$, since the change in volume is negligible ($\Delta V \approx 0$).

5 Favorable reactions have $\Delta G < 0$. We can deduce this from the second law and Equation 1 because the second law states that increasing entropy is favorable, and the equation has ΔG directly related to $-T\Delta S$.

Spontaneous reactions, ones that occur without a net addition of energy, have $\Delta G < 0$. They occur with energy to spare. Reactions with a negative ΔG are **exergonic** (energy *exits* the system); reactions with a positive ΔG are **endergonic**. Endergonic reactions occur only if energy is added. In the lab, energy is added in the form of heat; in the body, endergonic reactions are driven by reaction coupling to exergonic reactions (more on this later). Reactions with a negative ΔH are called **exothermic** and liberate heat. Most metabolic reactions are exothermic (which is how homeothermic organisms such as mammals maintain a constant body temperature). Reactions with a positive ΔH require an input of heat and are referred to as **endothermic**. (Thermodynamics will be discussed in more detail in *MCAT General Chemistry Review* and *MCAT Physics and Math Review*.)

The signs of thermodynamic quantities are assigned from the point of view of *the system*, not the surroundings or the universe. Thus, a negative ΔG means that the system goes to a lower free energy state, and a system will always move in the direction of the lowest free energy. As an analogy, visualize a spinning top as the system. What happens to the top? Does it spin faster and faster? No. It moves toward the lowest energy state. Let's expand the analogy, using an equation:

$$\text{motionless top} \rightarrow \text{spinning top}$$

Here the "reactant" is the motionless top, and the "product" is the spinning top. Which is lower: the free energy of product or reactant? The reactant. Is the reaction "spontaneous" as written? No; in fact, the reverse reaction is spontaneous. Therefore,

$$G_{\text{spinning}} > G_{\text{motionless}}$$

and thus,

$$\Delta G_{\text{reaction as written (motionless to spinning; left to right)}} > 0$$

So the reaction is nonspontaneous. In other words, *it requires energy input*, namely, energy from your muscles as you spin the top. [If the products in a reaction have more entropy than the reactants and the enthalpy (H) of the reactants and the products are the same, can the reaction occur spontaneously?[6]]

The value of ΔG depends on the concentrations of reactants and products, which can be variable in the body. Therefore, to compare reactions, chemists calculate a standard free energy change, denoted $\Delta G°$, with all reactants and products present at 1 *M* concentration. Under physiological conditions, however, the hydrogen ion concentration is far from 1 *M*, so biochemists use an even more standardized ΔG, with 1 *M* concentration for all solutes except H^+ and a pH of 7; this is denoted $\Delta G°'$.

$\Delta G°'$ is related to the equilibrium constant for a reaction by the following equation:

$$\textbf{Eq. 3} \quad \Delta G°' = -RT \ln K'_{eq}$$

[6] Yes. If $\Delta S > 0$ and $\Delta H = 0$, then according to the second law of thermodynamics, the reaction is spontaneous; see Equation 1.

where R is the gas constant (which would be given on the MCAT, along with the entire equation), and K'_{eq} is the ratio of products to reactants at equilibrium:

$$K'_{eq} = \frac{[C]_{eq}[D]_{eq}}{[A]_{eq}[B]_{eq}}$$

K'_{eq} is the ratio of products to reactants when enough time has passed for equilibrium to be reached. [When $K'_{eq} = 1$, what is $\Delta G°'$?[7]]

But what if we wanted to calculate ΔG for a reaction in the body? In this case, we need one more equation:

$$\textbf{Eq. 4} \quad \Delta G = \Delta G°' + RT \ln Q, \text{ where } Q = \frac{[C][D]}{[A][B]}$$

Here, Q is calculated using the actual concentrations of A, B, C, and D (for example, the concentrations in the cell). Equation 4 is simply a conversion from $\Delta G°'$ (the laboratory standard ΔG with initial concentrations at 1 M) to the real-life here-and-now ΔG. Note that if we put 1 M concentrations of A, B, C, and D into a beaker (at pH 7), we have recreated the laboratory standard initial set-up: $Q = 1$, so $\ln Q = 0$, which means $\Delta G = \Delta G°'$.

Remember that Q and K_{eq} are not the same. Q is the ratio of products to reactants in any given set-up; K_{eq} is the ratio *at equilibrium*. **Equilibrium** is defined as the point at which the rate of reaction in the forward direction equals the rate of reaction in the reverse direction. At equilibrium, there is constant product and reactant turnover as reactants form products and vice versa, but overall concentrations stay the same. Theoretically (given enough time), all reactant/product systems in a closed system will eventually reach this point.

While all reactions will eventually reach an equilibrium defined by the constant above, we can disturb this balance with the addition or removal of a reactant or product. This causes a change in Q but not K_{eq}, and the reaction will proceed in the direction necessary to reestablish equilibrium. (The shift to restore equilibrium is a demonstration of Le Châtelier's principle which will be discussed in further detail in *MCAT General Chemistry Review*.) Using this principle, a reaction which favors reactants at equilibrium can be driven to generate additional products (such strategies are employed frequently in cellular respiration).

- You are studying a particular reaction. You find the reaction in a book and read $\Delta G°'$ from a table. Can you calculate ΔG for this reaction in a living human being without any more information?[8]

[7] Equation 3 says that $\Delta G°' = 0$ when $K'_{eq} = 1$ since $\ln 1 = 0$. Note: for more information about MCAT Math, see *MCAT Physics and Math Review*.

[8] No. You need to know the concentrations of A, B, C, and D in the human cell. For example, $\Delta G°'$ might be +14.8 kcal/mol, indicating that the reaction is very unfavorable under standard conditions and has a $K < 1$, which means that at equilibrium there are more reactants than products. If, however, the ratio of reactants to products is made to be higher than that established at equilibrium, the Q for the reaction becomes less than K, and the forward reaction is spontaneous under these conditions, since ΔG will be less than zero (even though $\Delta G°'$ will not change). The significance of Q as an independent variable in Equation 4 is that it accounts for Le Châtelier's principle. If a system at equilibrium has reactants added to it, $Q < K$ and $\Delta G < 0$, so the high concentration of reactants will drive the reaction forward to reestablish equilibrium. If a system at equilibrium has products added to it, $Q > K$ and $\Delta G > 0$, so the high concentration of products will drive the reaction backward to reestablish equilibrium. All reactions at equilibrium will respond to these stresses in the same way, regardless of the sign of their $\Delta G°'$.

- How can ΔG be negative if $\Delta G°'$ is positive (which indicates that the reaction is unfavorable at standard conditions)?[9]
- Does K_{eq} indicate the rate at which a reaction will proceed?[10]
- When K_{eq} is large, which has lower free energy: products or reactants?[11]
- When Q is large, which has lower free energy: products or reactants?[12]
- Which direction, forward or backward, will be favored in a reaction if $\Delta G = 0$? (*Hint*: What does Equation 4 look like when $\Delta G = 0$?)[13]
- Radiolabeled chemicals are often used to trace constituents in biochemical reactions. The following reaction with $\Delta G = 0$ is in aqueous solution:

$$\mathrm{A} \rightleftharpoons \mathrm{B} + \mathrm{C}, \quad K_{eq} = \frac{[\mathrm{B}][\mathrm{C}]}{[\mathrm{A}]}$$

A small amount of radiolabeled B is added to the solution. After a period of time, where will the radiolabel most likely be found: in A, in B, or in both?[14]

In summary, then, there are two factors that determine whether a reaction will occur spontaneously (ΔG negative) in the cell:

1) The intrinsic properties of the reactants and products (K_{eq})
2) The concentrations of reactants and products ($RT \ln Q$)

(In the lab, there is third factor: temperature. If $\ln Q$ is negative and the temperature is high enough, ΔG will be negative, regardless of the value of $\Delta G°'$.)

[9] The reaction may be favorable ($\Delta G < 0$) if the ratio of the concentrations of reactants to products is sufficiently large to drive the reaction forward (that is, if RT ln Q is more negative than $\Delta G°'$ is positive, which would make their sum (which, by Equation 4, is ΔG) negative).

[10] K_{eq} indicates only the relative concentrations of reagents once equilibrium is reached, not the reaction rate (how fast equilibrium is reached).

[11] A large K_{eq} means that more products are present at equilibrium. Remember that equilibrium tends toward the lowest energy state. Hence, when K_{eq} is large, products have lower free energy than reactants.

[12] The size of Q says nothing about the properties of the reactants and products. Q is calculated from whatever the initial concentrations happen to be. It is K_{eq} that says something about the nature of reactants and products, since it describes their concentrations after equilibrium has been reached.

[13] If ΔG is 0, then neither the forward nor the reverse reaction is favored. Look at Equations 3 and 4. Note that when $\Delta G = 0$, Equation 4 reduces to Equation 3, and thus $Q = K_{eq}$ (which means Q at this moment is the same as K_{eq}, measured after the reaction system is allowed to reach equilibrium). When $Q = K_{eq}$, we are by definition at equilibrium. Understand and memorize the following: when $\Delta G = 0$, you are at equilibrium; forward reaction rate equals back reaction rate, and the net concentrations of reactants and products do not change.

[14] The reaction is in dynamic equilibrium where reactions are occurring in both directions, but at an equal rate. Because $\Delta G = 0$, we know that the forward reaction and the reverse reaction proceed at equal rates, even though we don't know the actual value. Therefore, after a period of time, the radiolabel will be present in both A and B.

Thermodynamics vs. Reaction Rates

The term *spontaneous* is used to describe a reaction system with $\Delta G < 0$. This can be misleading, since the common usage of the word *spontaneous* has a connotation of *rapid rate*; this is not what spontaneous means in the context of chemical reactions. For example, many reactions have a negative ΔG, indicating that they are "spontaneous" from a thermodynamic point of view, but they do not necessarily occur at a significant rate. Spontaneous means that a reaction is energetically favorable, *but it says nothing about the rate of reaction*.

Thermodynamics will tell you where a system starts and finishes but nothing about the path traveled to get there. The difference in free energy in a reaction is only a function of the nature of the reactants and products. Thus, ΔG does not depend on the pathway a reaction takes or the rate of reaction; it is only a measurement of the difference in free energy between reactants and products.

3.2 KINETICS AND ACTIVATION ENERGY (E_A)

The reason some spontaneous (i.e., *thermodynamically favorable*) reactions proceed very slowly or not at all is that a large amount of energy is required to get them going. For example, the burning of wood is spontaneous, but you can stare at a log all day and it won't burn. Some energy (heat) must be provided to kick-start the process.

The study of reaction rates is called **chemical kinetics**. All reactions proceed through a **transition state (TS)** that is unstable and takes a great deal of energy to produce. The transition state exists for a very, very short time, either moving forward to form products or breaking back down into reactants. The energy required to produce the transition state is called the **activation energy** ($\boldsymbol{E_a}$). This is the barrier that prevents many reactions from proceeding even though the ΔG for the reaction may be negative. The match you use to light your fireplace provides the activation energy for the reaction known as burning. It is the activation energy barrier that determines the kinetics of a reaction. [How would the rate of a spontaneous reaction be affected if the activation energy were lowered?[15]]

The concept of E_a is key to understanding the role of enzymes, so let's spend some time on it. To illustrate, take this reaction:

$$\text{Bob}_{\text{without a job}} + \text{job} \rightarrow \text{Bob}_{\text{with a job}}$$

Is this a favorable reaction, i.e., will the universe be better off, with less total (nervous) energy, if Bob gets the job? Will things settle down? Let's assume yes. However, between the two states (without/with), there is a temporary state, namely, $\text{Bob}_{\text{applying for job}}$. So the reaction will look this way:

$$\text{Bob}_{\text{without a job}} + \text{job} \rightarrow [\text{Bob}_{\text{applying for job}}]^{\ddagger} \rightarrow \text{Bob}_{\text{with a job}}$$

[15] The rate would be increased, since lowering E_a is tantamount to reducing the energy required to achieve the transition state. The more transition state products that are formed, the greater the amount of product produced, i.e., the more rapid the rate of reaction.

The middle term is the transition state, traditionally written in square brackets with a double-cross symbol: [TS]‡. The energy required for Bob to be job hunting is much higher than the energy of Bob with a job *or* Bob without a job. As a result, he may not go job hunting, even though he'd be happier in the long run if he did. In this model, we can describe the E_a as the energy necessary to get Bob to apply for a job.

A **catalyst** lowers the E_a of a reaction *without changing the* ΔG. The catalyst lowers the E_a by *stabilizing the transition state*, making its existence less thermodynamically unfavorable. The second important characteristic of a catalyst is that it is not consumed in the reaction; it is *regenerated* with each reaction cycle.

In our model, an example of a catalyst would be a career planning service (CPS). Adding a CPS won't make $\text{Bob}_{\text{without a job}}$ any happier or sadder, nor will it make $\text{Bob}_{\text{with a job}}$ happier or sadder. But it will make it much easier for Bob to move between the two states: without a job versus with a job. The traditional way to represent a reaction system like this is using a *reaction coordinate* graph, as shown in Figure 1. This is just a way to look at the energy of the reaction system as compared to the three possible states of the system: 1) reactants, 2) [TS]‡, and 3) products. The *x*-axis plots the physical progress of the reaction system (the "reaction coordinate"), and the *y*-axis plots energy.

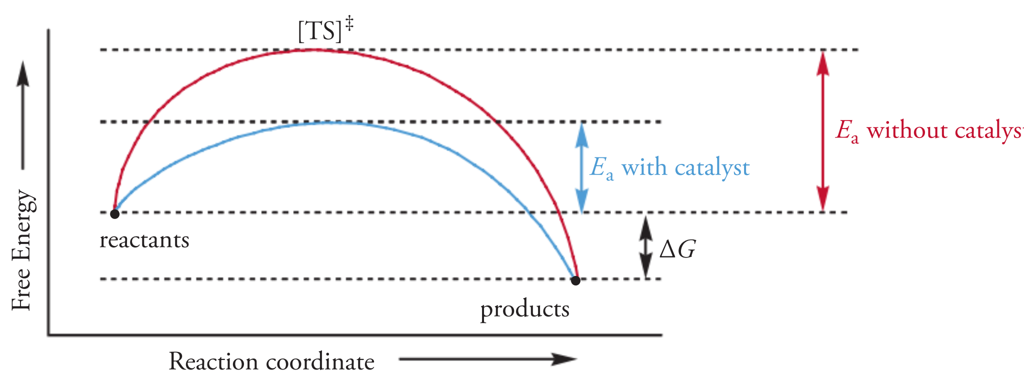


Figure 1 The reaction coordinate graph

Enzymes are biological catalysts. They increase the rate of a reaction by lowering the reaction's activation energy, but they *do not affect* ΔG between reactants and products. As catalysts, enzymes have a kinetic role, *not* a thermodynamic one. [Will an enzyme alter the concentration of reagents at equilibrium?[16]] Enzymes may alter the rate of a reaction enormously: a reaction that would take a hundred years to reach equilibrium without an enzyme may occur in just seconds with an enzyme. More information on enzymes can be found in Chapter 4.

[16] No. It will affect only the rate at which the reactants and products reach equilibrium.

3.3 OXIDATION AND REDUCTION

Energy Metabolism and the Definitions of Oxidation and Reduction

Where does the energy in foods come from? How do we make use of this energy? Why do we breathe? The answers begin with **photosynthesis**, the process by which plants store energy from the sun in the bond energy of carbohydrates. Plants are **photoautotrophs** because they use energy from light ("photo") to make their own ("auto") food. We are **chemoheterotrophs**, because we use the energy of chemicals ("chemo") produced by other ("hetero") living things, namely plants and other animals. Plants and animals store chemical energy in reduced molecules such as carbohydrates and fats. These reduced molecules are oxidized to produce CO_2 and ATP. The energy of ATP is used in turn to drive the energetically unfavorable reactions of the cell. That's the basic energetics of life; all the rest is detail.

In essence, the production and utilization of energy boil down to a series of oxidation/reduction reactions. **Oxidation** is a chemical term meaning the loss of electrons. **Reduction** means the opposite, the gain of electrons. Molecules can gain or lose electrons depending on the other atoms that they are bound to. There are three common ways to identify oxidation/reduction reactions on the MCAT, and it is important for you to know them:

Recognizing Oxidation Reactions:

1) gain of oxygen atoms
2) loss of hydrogen atoms
3) loss of electrons

Recognizing Reduction Reactions (just the opposite):

1) loss of oxygen atoms
2) gain of hydrogen atoms
3) gain of electrons

Though you should memorize this, it is not a subject worthy of philosophizing. If you can answer questions like the following, you're set: is changing CH_3CH_3 to $H_2C{=}CH_2$ an oxidation, a reduction, or neither?[17] What about changing Fe^{3+} to Fe^{2+}?[18] What about this: $O_2 \rightarrow H_2O$?[19]

[17] It's an oxidation, because hydrogens have been removed.

[18] It's a reduction, because an electron has been added.

[19] It's a reduction, because hydrogens have been added to the oxygen molecule.

You can also identify oxidation/reduction reactions visually by looking at the structure of the molecules. Is the formation of a disulfide bond (Figure 2) an oxidation or a reduction reaction?[20]

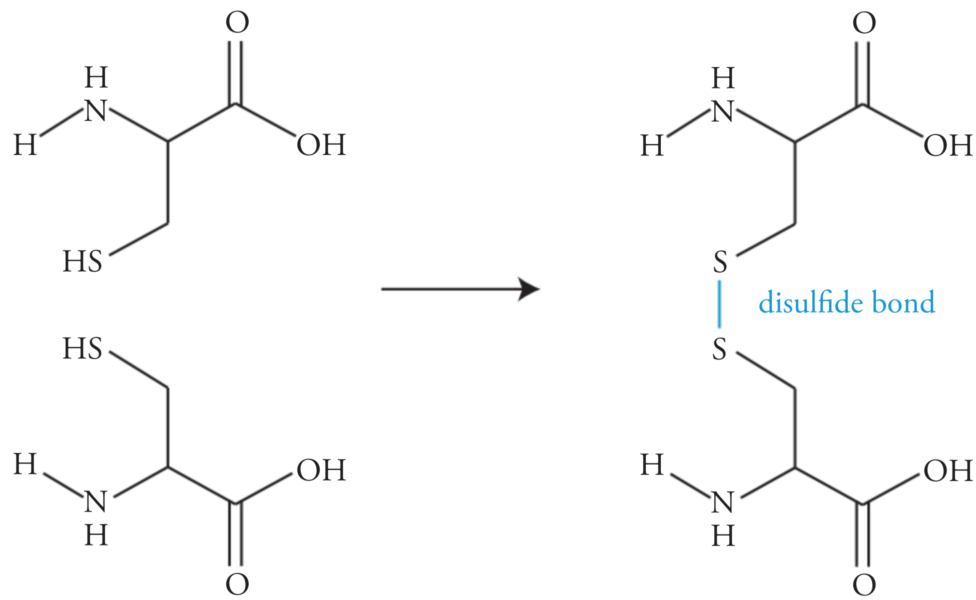


Figure 2 Formation of a disulfide bond

Here is one other important fact about oxidation and reduction: when one atom gets reduced, another one *must* be oxidized; hence the term ***redox pair***. As you study the process of glucose oxidation, you will see that each time an oxidation reaction occurs, a reduction reaction occurs too.

Catabolism is the process of breaking down molecules. The opposite is **anabolism**, which is "building-up" metabolism.[21] For example, the way we extract energy from glucose is by **oxidative catabolism**. We break down the glucose by oxidizing it. The stoichiometry of glucose oxidation looks like this:

$$C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$$

- What are the two members of the redox pair in this reaction?[22]

As we oxidize foods, we release the stored energy plants received from the sun. But we don't make use of that energy right away. Instead, we store it in the form of ATP. Alternatively, we can use the energy in ATP to generate storage molecules such as glycogen and fatty acids. Fatty acids are generated by successive reductions of a carbon chain, thus anabolic processes are generally reductive.

[20] It's an oxidation, because hydrogens have been removed.

[21] The mnemonics are *cata* = breakdown, as in catastrophe, and *ana* = buildup, sounds like "add-a." (Think of anabolic steroids, which weight-lifters use to bulk up.)

[22] The carbons in the sugar are oxidized (to CO_2), and oxygen is reduced (to H_2O).

3.4 ACIDS AND BASES

There are two important definitions of acids and bases you should be familiar with for biochemistry on the MCAT.

3.4

Brønsted-Lowry Acids and Bases

Brønsted and Lowry offered the following definitions:

Acids are proton (H^+) donors.
Bases are proton (H^+) acceptors.

While the often-seen hydroxide ions qualify as Brønsted-Lowry bases, many other compounds fit this definition as well. Since a Brønsted-Lowry base is any substance that is capable of accepting a proton, any anion or any neutral species with a lone pair of electrons can function as a base.

If we consider the reversible reaction below:

$$H_2CO_3 + H_2O \rightleftharpoons H_3O^+ + HCO_3^-$$

then according to the Brønsted-Lowry definition, H_2CO_3 and H_3O^+ are acids and HCO_3^- and H_2O are bases. The Brønsted-Lowry definition of acids and bases is the most important one for the MCAT.

Lewis Acids and Bases

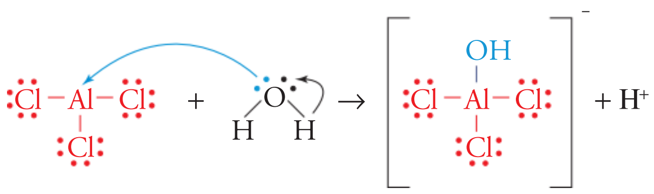
Lewis's definitions of acids and bases are broader:

Lewis acids are electron-pair acceptors.
Lewis bases are electron-pair donors.

If we consider the reversible reaction below:

$$AlCl_3 + H_2O \rightleftharpoons (AlCl_3OH)^- + H^+$$

then according to the Lewis definition, $AlCl_3$ and H^+ are acids because they accept electron pairs; H_2O and $(AlCl_3OH)^-$ are bases because they donate electron pairs. Lewis acid/base reactions frequently result in the formation of coordinate covalent bonds. For example, in the reaction above, water acts as a Lewis base, since it donates both of the electrons involved in the coordinate covalent bond between OH^- and $AlCl_3$. $AlCl_3$ acts as a Lewis acid, since it accepts the electrons involved in this bond.



The binding of an oxygen molecule to the iron atom in a heme group is a great biological example of a coordinate covalent bond formed between a Lewis acid and base:

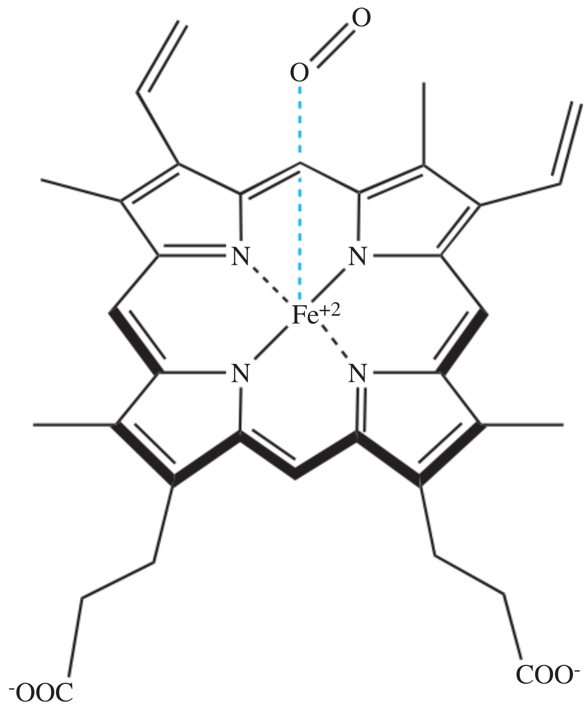


Figure 3 Example of a coordinate covalent bond

- Is oxygen the Lewis acid or Lewis base in the heme group?[23]

Conjugate Acids and Bases

When a Brønsted-Lowry acid donates an H^+, the remaining structure is called the **conjugate base** of the acid. Likewise, when a Brønsted-Lowry base bonds with an H^+ in solution, this new species is called the **conjugate acid** of the base. To illustrate these definitions, consider this reaction:

acid–base conjugates

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

acid–base conjugates

[23] Oxygen is donating a pair of electrons to the Fe^{2+} ion and is therefore the Lewis base.

Considering only the forward direction, NH_3 is the base and H_2O is the acid. The products are the conjugate acid and conjugate base of the reactants: NH_4^+ is the conjugate acid of NH_3, and OH^- is the conjugate base of H_2O:

conjugate base

acid $\dashrightarrow$

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

conjugate acid

base $\dashrightarrow$

Now consider the reverse reaction in which NH_4^+ is the acid and OH^- is the base. The conjugates are the same as for the forward reaction: NH_3 is the conjugate base of NH_4^+, and H_2O is the conjugate acid of OH^-:

conjugate base

$\dashleftarrow$ acid

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

conjugate acid

$\dashleftarrow$ base

The difference between a Brønsted-Lowry acid and its conjugate base is that the base is missing an H^+. The difference between a Brønsted-Lowry base and its conjugate acid is that the acid has an extra H^+.

forming conjugates:

$$\text{acid} \underset{+\,H^+}{\overset{-\,H^+}{\rightleftharpoons}} \text{base}$$

The Strengths of Acids and Bases

If we use HA to denote a generic acid, its dissociation in water has the form

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

The strength of the acid is directly related to how much the products are favored over the reactants. The equilibrium expression for this reaction is

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

This is written as K_a, rather than K_{eq}, to emphasize that this is the equilibrium expression for an acid-dissociation reaction. In fact, K_a is known as the **acid-ionization** (or **acid-dissociation**) **constant** of the acid (HA). We can rank the relative strengths of acids by comparing their K_a values: the larger the K_a value, the stronger the acid; the smaller the K_a value, the weaker the acid.

3.4

- Of the following acids, which one would dissociate to the greatest extent (in water)?[24]
  A) HCN (hydrocyanic acid), $K_a = 6.2 \times 10^{-10}$
  B) HNCO (cyanic acid), $K_a = 3.3 \times 10^{-4}$
  C) HClO (hypochlorous acid), $K_a = 2.9 \times 10^{-8}$
  D) HBrO (hypobromous acid), $K_a = 2.2 \times 10^{-9}$

We can apply the same ideas as above to identify the strength of *bases*. If we use B to denote a generic base, its dissolution in water has the form

$$B(aq) + H_2O(l) \rightleftharpoons HB^+(aq) + OH^-(aq)$$

Similar to acids, the strength of the base is directly related to how much the products are favored over the reactants. If we write the equilibrium constant for this reaction, we get:

$$K_b = \frac{[HB^+][OH^-]}{[B]}$$

This is written as K_b, rather than K_{eq}, to emphasize that this is the equilibrium expression for a base-dissociation reaction. In fact, K_b is known as the **base-ionization** (or **base-dissociation**) **constant**. We can rank the relative strengths of bases by comparing their K_b values: the larger the K_b value, the stronger the base; the smaller the K_b value, the weaker the base.

Amphoteric Substances

Take a look at the dissociation of carbonic acid (H_2CO_3), a weak acid:

$$H_2CO_3(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HCO_3^-(aq) \qquad (K_a = 4.5 \times 10^{-7})$$

The conjugate base of carbonic acid is HCO_3^-, which also has an ionizable proton. Carbonic acid is said to be **polyprotic**, because it has more than one proton to donate.

[24] **B.** The acid that would dissociate to the greatest extent would have the greatest K_a value. Of the choices given, HNCO (choice B) has the greatest K_a value.

Let's look at how the conjugate base of carbonic acid dissociates:

$$HCO_3^-(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CO_3^{2-}(aq) \quad (K_a = 4.8 \times 10^{-11})$$

In the first reaction, HCO_3^- acts as a base, but in the second reaction, it acts as an acid. Whenever a substance can act as either an acid or a base, we say that it is **amphoteric**. The conjugate base of a weak polyprotic acid is always amphoteric, because it can either donate or accept another proton. Also notice that HCO_3^- is a weaker acid than H_2CO_3; in general, every time a polyprotic acid donates a proton, the resulting species will be a weaker acid than its predecessor. Amino acids, which will be discussed in more detail in the next chapter, are all also amphoteric compounds and very important for the MCAT.

pH

The pH scale measures the concentration of H^+ (or H_3O^+) ions in a solution. Because the molarity of H^+ tends to be quite small and can vary over many orders of magnitude, the pH scale is logarithmic:

$$\text{pH} = -\log[H^+]$$

This formula implies that $[H^+] = 10^{-\text{pH}}$. Since $[H^+] = 10^{-7}$ *M* in pure water, the pH of water is 7. At 25°C, this defines a pH neutral solution. If $[H^+]$ is greater than 10^{-7} *M*, then the pH will be less than 7, and the solution is said to be acidic. If $[H^+]$ is less than 10^{-7} *M*, the pH will be greater than 7, and the solution is basic (or alkaline). Notice that a *low* pH means a *high* $[H^+]$ and the solution is *acidic*; a *high* pH means a *low* $[H^+]$ and the solution is *basic*.

pH > 7	basic solution
pH = 7	neutral solution
pH < 7	acidic solution

The range of the pH scale for most solutions falls between 0 and 14, but some strong acids and bases extend the scale past this range. For example, a 10 *M* solution of HCl will fully dissociate into H^+ and Cl^-. Therefore, the $[H^+] = 10$ *M*, and the pH = –1.

An alternate measurement expresses the acidity or basicity in terms of the hydroxide ion concentration, $[OH^-]$, by using pOH. The same formula applies for hydroxide ions as for hydrogen ions.

$$\text{pOH} = -\log[OH^-]$$

This formula implies that $[OH^-] = 10^{-\text{pOH}}$.

Acids and bases are inversely related: the greater the concentration of H^+ ions, the lower the concentration of OH^- ions, and vice versa. Since $[H^+][OH^-] = 10^{-14}$ at 25°C, the values of pH and pOH satisfy a special relationship at 25°C:

$$\text{pH} + \text{pOH} = 14$$

So, if you know the pOH of a solution, you can find the pH, and vice versa. For example, if the pH of a solution is 5, then the pOH must be 9. If the pOH of a solution is 2, then the pH must be 12.

3.4

Relationships Between Conjugates

pK_a and pK_b

The definitions of pH and pOH both involve a negative logarithm. In general, "p" of something is equal to the –log of that something. Therefore, the following definitions won't be surprising:

$$pK_a = -\log K_a$$

$$pK_b = -\log K_b$$

Because H^+ concentrations are generally very small and can vary over such a wide range, the pH scale gives us more convenient numbers to work with. The same is true for pK_a and pK_b. Remember that the larger the K_a value, the stronger the acid. Since "p" means "take the negative log of…," the *lower* the pK_a value, the stronger the acid. For example, lactic acid has a K_a of 1.3×10^{-4}, and uric acid has a K_a of 2.5×10^{-6}. Since the K_a of lactic acid is larger than that of uric acid, more molecules of lactic acid than uric acid will dissociate into ions in aqueous solution. In other words, lactic acid is stronger than uric acid. The pK_a of lactic acid is 3.9, and the pK_a of uric acid is 5.6. **The acid with the lower pK_a value is the stronger acid. The same logic applies to pK_b: the lower the pK_b value, the stronger the base.** Memorize this!

- Of the following liquids, which one contains the lowest concentration of H_3O^+ ions?[25]
  A) Lemon juice (pH = 2.3)
  B) Blood (pH = 7.4)
  C) Seawater (pH = 8.5)
  D) Coffee (pH = 5.1)

- Which of the following compounds is the least acidic?[26]
  A) CH_3COOH, acetic acid (pK_a = 4.76)
  B) H_2CO_3, carbonic acid (pK_a = 6.35)
  C) H_3PO_4, phosphoric acid (pK_a = 2.15)
  D) HCO_3^-, bicarbonate (pK_a = 10.33)

[25] **C.** Since pH = – log [H_3O^+], we know that [H_3O^+] = 1/10 pH. This fraction is smallest when the pH is greatest. Of the choices given, seawater has the highest pH.

[26] **D**. The compound with the highest pK_a is the least acidic. Bicarbonate has the highest pK_a of all the answer choices.

Buffer Solutions

A **buffer** is a solution that resists changing pH when a small amount of acid or base is added. The buffering capacity comes from the presence of a weak acid and its conjugate base (or a weak base and its conjugate acid) in roughly equal concentrations.

Buffers are extremely important because nearly every biological process in the human body is pH-dependent. The H^+ ion commonly functions as a reactant, product, or catalyst in chemical reactions in metabolic pathways. In addition, there are many sources of acids and bases in the body that can cause significant changes in the pH if it weren't for our buffer systems.

The most important buffer system in our blood plasma (and for the MCAT) is the bicarbonate buffer system. This buffer consists of carbonic acid (H_2CO_3) and its conjugate base, bicarbonate (HCO_3^-):

Reaction 1 $\quad H_2CO_3 \rightarrow H^+ + HCO_3^-$

This buffer system is particularly complex because of how carbonic acid is formed in the body. During cellular respiration, an activity that is constantly occurring in our body, our cells produce carbon dioxide (CO_2) as a byproduct. The carbon dioxide can then react in a reversible fashion with water to form carbonic acid:

Reaction 2 $\quad CO_2 + H_2O \rightarrow H_2CO_3$

To understand how this buffer system works, we can consider the following scenario. Let's say you go on a run, causing your muscle tissue to produce lactic acid. The lactic acid would seek to increase the concentration of H^+ ions in your body and thus decrease the pH. As discussed earlier, a drop in pH is problematic; it can severely impact our metabolic processes. However, when the lactic acid produces H^+ ions, Reaction 1 shifts to the left by Le Châtelier's principle, reducing the amount of free H^+ ions. While this shift does not completely prevent the pH from falling, it significantly reduces the degree to which the pH falls.

Chapter 3 Summary

- ΔG, the Gibbs free energy, is the amount of energy in a reaction available to do chemical work.

- For a reaction under any set of conditions, $\Delta G = \Delta H - T\Delta S$.

- If $\Delta G < 0$, the reaction is spontaneous in the forward direction. If $\Delta G > 0$, the reaction is nonspontaneous in the forward direction. If $\Delta G = 0$, the reaction is at equilibrium.

- Kinetics is the study of how quickly a reaction occurs, but it does not determine *whether or not* a reaction will occur.

- Activation energy (E_a) is the minimum energy required to start a reaction and decreases in the presence of a catalyst, thereby increasing the reaction rate.

- Acids are proton donors and electron acceptors; bases are proton acceptors and electron donors.

- The higher the K_a (lower the pK_a), the stronger the acid. The higher the K_b (lower the pK_b), the stronger the base.

- Amphoteric substances may act as either acids or bases.

- pH = $-\log[H^+]$. For a concentration of H^+ given in a 10^{-x} M notation, simply take the negative exponent to find the pH. The same is true for the relationship between $[OH^-]$ and pOH, K_a and pK_a, and K_b and pK_b.

- Buffers resist pH change upon the addition of a small amount of acid or base. The key buffer system in the body is the bicarbonate buffer system.

CHAPTER 3 FREESTANDING PRACTICE QUESTIONS

1. Which of the following best describes the function of enzymes?

A. By decreasing the E_a, enzymes increase both the reaction rate and the total amount of product formed.
B. By making the ΔG of the reaction more negative, enzymes increase the amount of product formed per unit time.
C. By decreasing the energy of the transition state, enzymes increase the amount of product formed per unit time.
D. By making the ΔG of the reaction more positive, enzymes increase the total amount of product formed.

2. Malate dehydrogenase is a key enzyme in TCA cycle, which catalyzes the following reaction, which is unfavorable under standard conditions:

$$\mathbf{C_4H_6O_5 + NAD^+ \rightarrow C_4H_4O_5 + NADH + H^+}$$

Which of the following corresponds to the compound that is most likely to be oxidized under standard conditions?

A. $C_4H_6O_5$
B. NAD^+
C. $C_4H_4O_5$
D. NADH

3. All of the following are examples of oxidation-reduction reactions EXCEPT:

I. $C_3H_7O_6P + NAD^+ + P_i \rightarrow C_3H_8O_{10}P_2 + NADH + H^+$
II. $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$
III. $C_6H_{14}O_{12}P_2 \rightarrow 2\ C_3H_7O_6P$

A. II only
B. III only
C. I and II only
D. I and III only

4. An acidic Glu residue in a protein (neutral in its protonated form) has a pK_a value of 2.3 in the wild-type protein and is found near a neutral Ile residue. What will be the effect on the pK_a of the Glu residue if a mutation substitutes a positively charged Lys residue for the Ile residue?

A. The pK_a will decrease due to favorable ionic interactions between the deprotonated Glu and Lys residues.
B. The pK_a will decrease due to unfavorable ionic interactions between the deprotonated Glu and Lys residues.
C. The pK_a will increase due to favorable ionic interactions between the deprotonated Glu and Lys residues.
D. The pK_a will increase due to unfavorable ionic interactions between the deprotonated Glu and Lys residues.

5. Which of the following best orders the relative basicity of the side chains of Glu (pK_a = 4.1), Cys (pK_a = 8.3), and Lys (pK_a = 10.8)?

A. Cys > Glu > Lys
B. Lys > Glu > Cys
C. Lys > Cys > Glu
D. Glu > Cys > Lys

6. The cells in your body are constantly undergoing cellular respiration, producing CO_2 as a byproduct. What would happen to the pH of your blood if you were to hold your breath?

A. It would increase.
B. It would be the same.
C. It would decrease.
D. It cannot be determined with the information provided.

CHAPTER 3 PRACTICE PASSAGE

The complexity of hemoglobin's function is exemplified by its chemical structure. Multiple factors are believed to affect the kinetics of oxygen binding to the molecule's active sites. It has been proposed that the binding of oxygen (Figure 1) reduces strain on the heme protein superstructure by counterbalancing the pull of the proximal imidazole. Similarly, increasing basicity of the proximal imidazole (e.g., through loss of a hydrogen atom) is believed to exert additional strain on the heme protein in the deoxygenated state by inducing a dome-shaped molecular structure. Because the binding of oxygen relieves this strain, increasing basicity of the proximal imidazole increases the affinity of the heme for oxygen. The hydrophobic pocket created by hydrocarbon-like residues from adjacent heme proteins is also believed to facilitate oxygenation, though it has been proposed steric hindrance may provide an antagonistic effect. The distal imidazole shown below may inhibit dissociation by stabilizing the oxygen molecule in place.

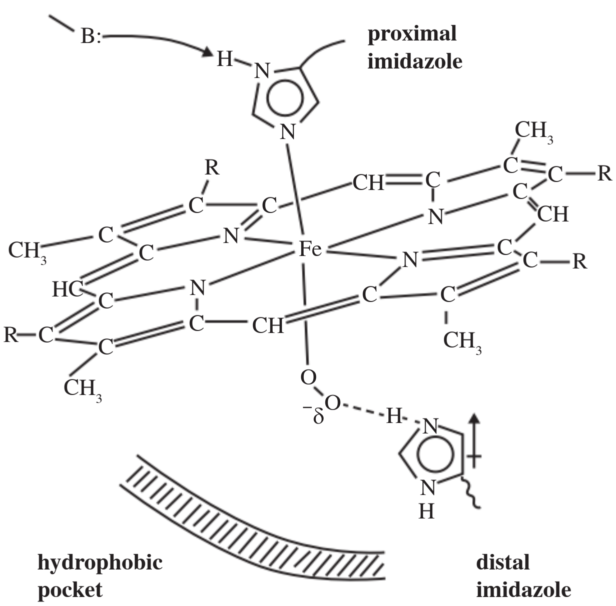


Figure 1 Summary schematic of selected stereoelectronic factors in oxygen affinity of heme

The net effect of these and other factors affecting oxygen dissociation from hemoglobin can be expressed kinetically:

Equation 1 $$\text{Heme} + O_2 \underset{k}{\overset{k'}{\rightleftharpoons}} \text{Heme-}O_2$$

A single heme molecule will associate with O_2 to form a heme-O_2 complex with rate constant k'; in a reverse reaction, oxygen will dissociate from the heme-O_2 complex with rate constant k. One study examined the kinetics of association of oxygen at different oxygen-bound states as well as the implied equilibrium constants (K_{O_2}) for each stage of oxygen association (Table 1).

Table 1 Kinetic Data Pertaining To Oxygen Association With Hemoglobin

Heme	k' (M^{-1} sec^{-1} $\times 10^{-7}$)	k (sec^{-1})	K'_{O_2} ($M^{-1} \times 10^{-6}$)
Hb	0.20	1079	0.0019
Hb(O_2)	0.39	245	0.016
Hb$(O_2)_2$	0.35	30	0.12
Hb$(O_2)_3$	4.00	47	0.85

Adapted from Chang, C. K., and Traylor, T. G. *Kinetics of oxygen and carbon monoxide binding to synthetic analogs of the myoglobin and hemoglobin active sites;* Ip, S. H. C.

1. Based on information presented in the passage, which of the following most likely describes the outcome of replacing the distal imidazole with a nonpolar amino acid residue?

A. The bond between iron and oxygen would be stabilized, decreasing the rate at which oxygen dissociates from hemoglobin.
B. The bond between iron and oxygen would be stabilized, increasing the rate at which oxygen dissociates from hemoglobin.
C. The bond between iron and oxygen would be destabilized, decreasing the rate at which oxygen dissociates from hemoglobin.
D. The bond between iron and oxygen would be destabilized, increasing the rate at which oxygen dissociates from hemoglobin.

2. Which of the following amino acids would likely NOT be found in the hydrophobic pocket of hemoglobin?

A. F
B. Y
C. L
D. V

3. What is the most likely effect of the addition of acid to the solution in which hemoglobin is suspended?

A. The structure of deoxyhemoglobin would become relatively less strained, reducing affinity for oxygen.
B. The structure of deoxyhemoglobin would become relatively less strained, increasing affinity for oxygen.
C. The structure of deoxyhemoglobin would become relatively more strained, reducing affinity for oxygen.
D. The structure of deoxyhemoglobin would become relatively more strained, increasing affinity for oxygen.

4. Which of the following would be most likely to bind oxygen the fastest in the pulmonary vasculature?

A. Hb
B. $Hb(O_2)$
C. $Hb(O_2)_2$
D. $Hb(O_2)_3$

SOLUTIONS TO CHAPTER 3 FREESTANDING PRACTICE QUESTIONS

1. **C** Enzymes are biological catalysts that affect the kinetics of biological reactions. By lowering the energy of the transition state, enzymes decrease the E_a of biological reactions and increase the amount of product formed per unit time (choice C is correct). ΔG is a thermodynamic quantity and will be unaffected by the addition of an enzyme (choices B and D are wrong). The total amount of product formed once the reaction has reached equilibrium is determined by thermodynamic factors, whereas the amount of product formed in a given time period is determined by kinetic factors (choice A is wrong).

2. **D** Given that the question stem indicates that the reaction is unfavorable under standard conditions, it is likely to proceed in the reverse direction under standard conditions. NADH loses a hydrogen atom in forming NAD^+ in the reverse reaction and is thereby oxidized (choice D is correct). Both $C_4H_6O_5$ and NAD^+ are more likely to be the product, rather than the reactant, of this redox reaction under standard conditions (choices A and B are wrong). Because $C_4H_4O_5$ gains hydrogen atoms in generating $C_4H_6O_5$, it is reduced, not oxidized (choice C is wrong).

3. **B** Since all of the items appear in exactly two answer choices, start by evaluating the shortest item, Item III. Item III is NOT an oxidation reduction reaction and is a correct response. This reaction simply splits $C_6H_{14}O_{12}P_2$ into two, three-carbon phosphorylated molecules, while maintaining the O to H ratio (choices A and C can be eliminated). Since Item II does not appear in any of the remaining choices, it must be an oxidation-reduction reaction (i.e., it is a false statement) and you can evaluate Item I. Item I is an oxidation-reduction reaction and is a false statement: NAD^+ gains a hydrogen atom and is reduced to NADH, and $C_3H_7O_6P$ must therefore be oxidized (choice D can be eliminated and choice B is correct). Note that Item II is in fact an oxidation-reduction reaction (a false statement): O_2 is in its standard state, giving it a zero oxidation state. It is reduced to have a –2 oxidation state in the products. The O to H ratio is substantially increased in generating CO_2 from $C_6H_{12}O_6$, which indicates $C_6H_{12}O_6$ has been oxidized.

4. **A** The substitution of a positively charged lysine for a neutral isoleucine would help stabilize Glu in its deprotonated, negatively-charged state (choices B and D can be eliminated). Because it would be more stable, it is more likely to give up its proton (i.e., it becomes more acidic), and the pK_a would decrease (choice C is wrong, and choice A is correct).

5. **C** pK_a is a measure of acid strength. The lower the pK_a, the more acidic the functional group. The question, however, is asking about basicity. Logically, the more acidic the functional group, the less basic it is. Therefore, the most basic group is the one with the highest pK_a, and the least basic is the one with lowest pK_a. Of the three amino acids, the side chain of Lys has the highest pK_a (choices A and D can be eliminated), and the side chain of Glu has the lowest pK_a (choice B can be eliminated, and choice C is correct).

6. **C** When you hold your breath, the CO_2 concentration in your blood increases. The increase in CO_2 will increase the formation of carbonic acid, H_2CO_3, in your blood, which will decrease the pH (choice C is correct; choices A, B, and D are wrong).

SOLUTIONS TO CHAPTER 3 PRACTICE PASSAGE

1. **D** In its protonated form, the distal imidazole forms a hydrogen bond with the oxygen molecule, stabilizing its position with respect to the iron. Introducing a nonpolar amino acid residue in place of this imidazole would remove the hydrogen bond, destabilizing the interaction between iron and oxygen (choices A and B are wrong). If the interaction between iron and oxygen is destabilized, this would likely result in an increase in the rate at which oxygen dissociates from heme (choice D is correct, and choice C is wrong).

2. **B** Highlight the word NOT and write "A B C D" on the noteboard. Evaluate each answer choice, writing "Y" if it is a hydrophobic amino acid that would be found in the pocket, or "N" if it is not; then choose the answer that stands out. F is phenylalanine, a hydrophobic amino acid (write "Y" next to "A" on the noteboard). Y is tyrosine, a polar amino acid (write "N" next to "B" on the noteboard). L is leucine and V is valine, both hydrophobic amino acids (write "Y" next to "C" and "D" on the noteboard). Since choice B stands out with an "N" instead of a "Y," it is the correct answer choice.

3. **A** Increasing the acidity of the buffer in which hemoglobin is suspended would effectively *decrease* the basicity of the proximal imidazole by maximizing the chance that H^+ will remain bound to the five-member ring. As a result, the proximal imidazole would exert less strain on the deoxygenated state of hemoglobin (choices C and D can be eliminated). According to the passage, increased basicity of the proximal imidazole would increase the affinity of the heme for oxygen; therefore, decreased basicity would decrease the affinity for oxygen (choice B can be eliminated, and choice A is correct). Indeed, while there are numerous effects of increasing acidity (decreasing pH) on the heme molecule, the commonly depicted oxygen dissociation curve below demonstrates a relative decrease in heme affinity for oxygen with increasing acidity; higher pO_2 is required for the same oxyhemoglobin saturation.